Place Value and Powers of 10

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS digit *(mati)* · place value *(uara tū)* · million · power of 10 · exponent *(taupū)* · expanded form

A Place value in large numbers ●○○

Remember: The place of a digit tells you its value. In 3,482,917 the 4 is in the hundred thousands place, so it is worth 400,000.

1. Write the value of the underlined digit.

a) $\underline{4},582$ _____ **b)** $3\underline{6},907$ _____ **c)** $\underline{2}18,450$ _____

d) $\underline{5},760,312$ _____ **e)** $90\underline{4},615$ _____ **f)** $1,\underline{2}83,590$ _____

g) $72,\underline{8}46$ _____ **h)** $6,0\underline{4}5,200$ _____ **i)** $3,91\underline{7},484$ _____

B Powers of 10 ●○○

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

2. Write each power of 10 as an ordinary number.

a) 10^2 = _____ **b)** 10^3 = _____ **c)** 10^1 = _____

d) 10^5 = _____ **e)** 10^4 = _____ **f)** 10^6 = _____

3. Write each number as a power of 10.

a) 1,000 = _____ **b)** 100 = _____ **c)** 100,000 = _____

d) 10 = _____ **e)** 1,000,000 = _____ **f)** 10,000 = _____

C Expanded form ●●○

Remember: Each digit is multiplied by the power of 10 for its place: $4{,}205 = 4 \times 10^3 + 2 \times 10^2 + 5 \times 10^0$. A place that holds a zero is left out.

4. Write as an ordinary number.

a) $4 \times 10^1 + 7 \times 10^0$

b) $3 \times 10^2 + 6 \times 10^1 + 2 \times 10^0$

c) $5 \times 10^3 + 8 \times 10^1$

d) $2 \times 10^4 + 4 \times 10^2 + 9 \times 10^0$

e) $3 \times 10^5 + 7 \times 10^2$

f) $6 \times 10^6 + 1 \times 10^3 + 5 \times 10^1$

5. Write each number in expanded form, using powers of 10.

a) 58

b) 731

c) 4,206

d) 90,350

e) 108,000

f) 3,050,400

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D Comparing and ordering ●●○

Remember: Compare the digits from the left. A number with more digits is always larger.

6. Write <, > or = in each box.

a) $4{,}520 \square 4{,}502$ **b)** $98{,}765 \square 100{,}234$ **c)** $360{,}000 \square 306{,}000$

d) $10^3 \square 1{,}000$ **e)** $9{,}999 \square 10^4$ **f)** $10^5 \square 50{,}000$

g) $1{,}200{,}000 \square 10^6$ **h)** $10^2 \square 10^3$

7. Write each list in order, from smallest to largest.

a) 4,310, 4,130, 4,031, 4,301 _____

b) 65,200, 62,500, 65,020, 56,200 _____

c) 1,040,000, 1,004,000, 1,400,000, 140,000 _____

E Multiplying and dividing by 10, 100 and 1,000 ●●○

Watch out: Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

8. Work out.

a) 45×100 = _____ **b)** 3.7×10 = _____ **c)** $0.62 \times 1{,}000$ = _____ **d)** 208×10 = _____

e) $4{,}800 \div 100$ = _____ **f)** $56 \div 10$ = _____ **g)** $7 \div 100$ = _____ **h)** $930 \div 1{,}000$ = _____

i) 1.05×100 = _____ **j)** 0.4×10 = _____ **k)** $2.5 \div 10$ = _____ **l)** $69 \div 100$ = _____

★ Extension

9. How many times larger is the value of the 7 in 7,300 than the value of the 7 in 73?

10. Find n if $10^3 \times 10^2 = 10^n$.

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NUMBER

Place Value and Powers of 10

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KEY WORDS digit *(mati)* · place value *(uara tū)* · million · power of 10 · exponent *(taupū)* · expanded form

A Place value in large numbers ●○○

Remember: The place of a digit tells you its value. In 3,482,917 the 4 is in the hundred thousands place, so it is worth 400,000.

1. Write the value of the underlined digit.

a) $\underline{7},265$ _____ **b)** $5\underline{8},304$ _____ **c)** $\underline{4}31,720$ _____

d) $\underline{8},250,961$ _____ **e)** $60\underline{3},847$ _____ **f)** $2,\underline{4}91,350$ _____

g) $96,\underline{5}18$ _____ **h)** $7,0\underline{3}2,600$ _____ **i)** $5,82\underline{6},193$ _____

B Powers of 10 ●○○

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

2. Write each power of 10 as an ordinary number.

a) 10^3 = _____ **b)** 10^2 = _____ **c)** 10^4 = _____

d) 10^1 = _____ **e)** 10^6 = _____ **f)** 10^5 = _____

3. Write each number as a power of 10.

a) 100 = _____ **b)** 10,000 = _____ **c)** 1,000 = _____

d) 10 = _____ **e)** 100,000 = _____ **f)** 1,000,000 = _____

C Expanded form ●●○

Remember: Each digit is multiplied by the power of 10 for its place: $4{,}205 = 4 \times 10^3 + 2 \times 10^2 + 5 \times 10^0$. A place that holds a zero is left out.

4. Write as an ordinary number.

a) $8 \times 10^1 + 3 \times 10^0$

b) $5 \times 10^2 + 9 \times 10^1 + 4 \times 10^0$

c) $7 \times 10^3 + 2 \times 10^1$

d) $4 \times 10^4 + 6 \times 10^2 + 7 \times 10^0$

e) $5 \times 10^5 + 3 \times 10^2$

f) $2 \times 10^6 + 4 \times 10^3 + 7 \times 10^1$

5. Write each number in expanded form, using powers of 10.

a) 74

b) 926

c) 5,308

d) 60,720

e) 205,000

f) 4,070,100

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D Comparing and ordering ●●○

Remember: Compare the digits from the left. A number with more digits is always larger.

6. Write <, > or = in each box.

a) $6{,}730 \square 6{,}703$ **b)** $87{,}654 \square 100{,}123$ **c)** $540{,}000 \square 504{,}000$

d) $10^2 \square 100$ **e)** $999 \square 10^3$ **f)** $10^4 \square 40{,}000$

g) $1{,}500{,}000 \square 10^6$ **h)** $10^5 \square 10^4$

7. Write each list in order, from smallest to largest.

a) 7,520, 7,250, 7,052, 7,502 _____

b) 83,400, 84,300, 83,040, 38,400 _____

c) 2,060,000, 2,006,000, 2,600,000, 260,000 _____

E Multiplying and dividing by 10, 100 and 1,000 ●●○

Watch out: Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

8. Work out.

a) 63×100 = _____ **b)** 5.2×10 = _____ **c)** $0.48 \times 1{,}000$ = _____ **d)** 307×10 = _____

e) $6{,}500 \div 100$ = _____ **f)** $84 \div 10$ = _____ **g)** $9 \div 100$ = _____ **h)** $710 \div 1{,}000$ = _____

i) 2.06×100 = _____ **j)** 0.7×10 = _____ **k)** $3.5 \div 10$ = _____ **l)** $48 \div 100$ = _____

Extension

9. How many times larger is the value of the 5 in 5,200 than the value of the 5 in 52?

10. Find n if $10^4 \times 10^2 = 10^n$.

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